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Biology

FULL LESSON NOTES

COMPLETE · LESSONS 1-96

UPPER SIXTH

GCE A Level · MINESEC Progression

10 chapters · every lesson of the official MINESEC progression sheet, with labelled diagrams

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How to use these notes

These notes follow the official **MINESEC progression sheet** lesson by lesson. Each lesson has its own section, numbered exactly as in the progression, so you always know where you are in the syllabus.

Feature	What to do with it
Learning objectives (green box)	Read them first: they tell you what you must be able to do by the end of the lesson.
Lesson notes and Key Points (blue box)	Learn the definitions and formulas in the Key Points by heart.
Exam tips (yellow box)	Common mistakes and how GCE examiners expect answers to be set out.
Worked examples	Cover the solution, try the question yourself, then compare line by line.
Practice questions	Do them without looking at the answers. Mark your work and redo any you got wrong.
Review exercise	Mixed exam-style questions to test yourself on several lessons at once.

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Chapter 1: Reproduction

Lesson	Title (MINESEC progression)
1	Concepts and types of asexual reproduction
2	Forms of asexual reproduction in animals
3	Sexual reproduction; formation of pollen and ovules
4	Pollination and fertilisation
5	Seed formation and dispersal
6	Germination
7	The male reproductive system
8	Hormonal control of the testes and spermatogenesis
9	The female reproductive system
10	Hormonal control of the ovaries
11	Menstruation and oogenesis
12	Fertilisation and development of the embryo
13	The birth process (parturition)
14	Reproduction (integration activity)
15	Evaluation
16	Correction and remediation
17	Alternation of generations
18	Metagenesis
19	Methods of birth control
20	Growth curves and growth patterns
21	Hormonal control of growth

Lesson 1: Concepts and types of asexual reproduction

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ define asexual reproduction and list its types
- ✔ state the advantages and disadvantages of asexual reproduction

Lesson notes

Reproduction is the production of new individuals of the same species. It ensures the survival of the species.

Asexual reproduction involves one parent and no fusion of gametes; offspring are produced by mitosis and are genetically identical to the parent (a **clone**).

Type	Mechanism	Examples
Binary fission	the cell divides into two equal parts	bacteria, Amoeba, Paramecium
Budding	a small outgrowth (bud) grows and separates	yeast, Hydra
Fragmentation	the body breaks into pieces, each regrows	Spirogyra, flatworms (Planaria)

Type	Mechanism	Examples
Spore formation	many small spores made by mitosis, dispersed	Mucor, ferns, mosses
Vegetative propagation	a part of the plant grows into a new plant	see below
Parthenogenesis	development of an egg without fertilisation	aphids, some bees (drones), some lizards

Natural vegetative propagation in plants: stem tubers (Irish potato), rhizomes (ginger, banana suckers), corms (cocoyam), bulbs (onion), runners/stolons (strawberry, grass), suckers (banana, plantain, pineapple), leaf plantlets (Bryophyllum).

Artificial propagation: cuttings (cassava, sugar cane, hibiscus), grafting (citrus, mango), layering, budding, tissue culture (micropropagation of banana, pineapple).

Advantages of asexual reproduction	Disadvantages
only one parent needed	no genetic variation: all offspring equally vulnerable to disease or environmental change
fast; large numbers of offspring	overcrowding and competition near the parent
good characteristics of the parent are preserved	harmful mutations are passed on
offspring well adapted to the same environment	poor dispersal (in many cases)

Practice Questions: Lesson 1

1. Name the type of asexual reproduction in (a) Amoeba (b) yeast (c) cassava stem cuttings.
2. Give one advantage and one disadvantage of asexual reproduction.

Answers: *1. (a) binary fission (b) budding (c) vegetative propagation 2. advantage: fast, no mate needed, offspring keep good features; disadvantage: no variation, so all may be killed by the same disease or change*

Lesson 2: Forms of asexual reproduction in animals

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ describe forms of asexual reproduction in animals

Lesson notes

- **Binary fission (Amoeba):** the nucleus divides by mitosis, then the cytoplasm divides into two daughter cells.
- **Multiple fission (Plasmodium in red cells):** the nucleus divides many times, then the cytoplasm divides into many merozoites.
- **Budding (Hydra):** a bud grows from the body wall, develops tentacles and a mouth, then detaches.
- **Fragmentation and regeneration (Planaria, starfish):** each piece regrows the missing parts.
- **Parthenogenesis:** in honey bees, unfertilised eggs develop into haploid males (drones); fertilised eggs become females (workers and queens).

Practice Questions: Lesson 2

1. What is parthenogenesis? Give an example.

Answers: *1. development of an egg without fertilisation (e.g. aphids, drone bees)*

Lesson 3: Sexual reproduction; formation of pollen and ovules

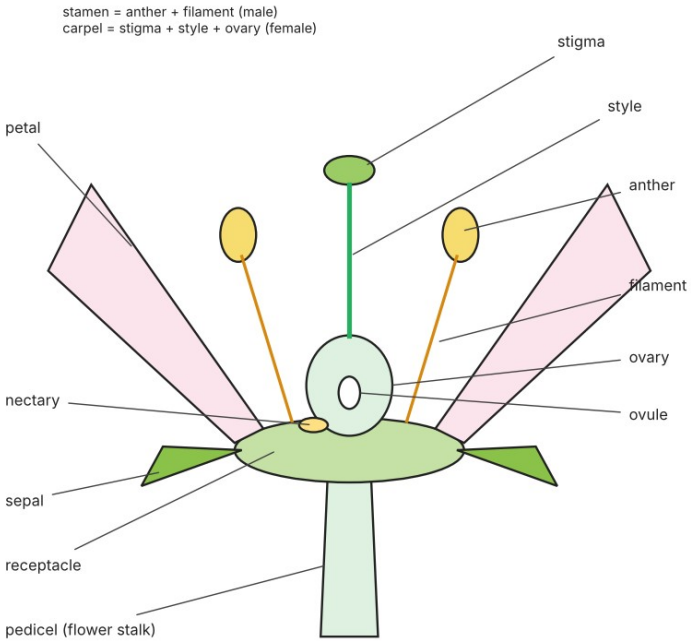
LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ describe the structure of a flower
- ✔ describe the formation of pollen grains and ovules

Lesson notes

Sexual reproduction involves the fusion of two haploid gametes (fertilisation) to form a diploid **zygote**. Gametes are made by meiosis, so offspring show **genetic variation**.



Formation of pollen (in the anther): diploid pollen mother cells in the pollen sacs divide by **meiosis** to form tetrads of four haploid microspores. Each microspore develops a tough wall (exine) and divides by mitosis into a **tube nucleus** and a **generative nucleus**; the generative nucleus later divides to give **two male nuclei (gametes)**.

Formation of the ovule and embryo sac (in the ovary): a megaspore mother cell in the ovule divides by meiosis into four megaspores; three degenerate. The surviving megaspore divides by mitosis three times to form the **embryo sac** with 8 haploid nuclei: an **egg cell** and 2 synergids at the micropyle end, 2 **polar nuclei** in the centre, 3 antipodal cells at the other end. The ovule is surrounded by integuments with a small opening, the **micropyle**.

Practice Questions: Lesson 3

1. How many nuclei are in a mature embryo sac?
2. State the function of the tapetum.

Answers: *1. eight 2. nourishes the developing pollen grains*

Lesson 4: Pollination and fertilisation

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ compare insect- and wind-pollinated flowers
- ✔ describe double fertilisation in flowering plants

Lesson notes

Pollination is the transfer of pollen from an anther to a stigma. **Self-pollination:** to the same flower or plant. **Cross-pollination:** to a different plant of the same species (more variation).

Feature	Insect-pollinated flowers	Wind-pollinated flowers
Petals	large, brightly coloured, scented	small, dull green, or absent
Nectar	present	absent
Pollen	small amounts, large, sticky or spiky	huge amounts, small, light, smooth
Anthers	inside the flower, firmly attached	hanging outside on long filaments
Stigma	inside, sticky	large, feathery, hanging outside
Examples	hibiscus, Crotalaria, sunflower	maize, rice, grasses

Mechanisms that favour cross-pollination: stamens and stigma mature at different times (protandry, protogyny); male and female flowers on different plants (dioecious, e.g. pawpaw); self-incompatibility; flower structure.

Fertilisation (double fertilisation in flowering plants):

- 1. A compatible pollen grain germinates on the stigma; a **pollen tube** grows down the style (digesting tissues with enzymes), guided by chemicals, and enters the ovule through the micropyle.
- 2. The tube nucleus leads; the two male nuclei follow.
- 3. One male nucleus fuses with the **egg cell** → diploid **zygote** (→ embryo).
- 4. The other fuses with the two **polar nuclei** → triploid **primary endosperm nucleus** (→ endosperm, food store).

Practice Questions: Lesson 4

1. Give three differences between wind- and insect-pollinated flowers.
2. What are the products of double fertilisation?

Answers: *1. wind: small dull flowers, no scent or nectar, long feathery stigmas and exposed anthers, light smooth pollen; insect: large bright scented flowers with nectar, sticky stigmas, sticky spiky pollen 2. a diploid zygote (embryo) and a triploid endosperm nucleus*

Lesson 5: Seed formation and dispersal

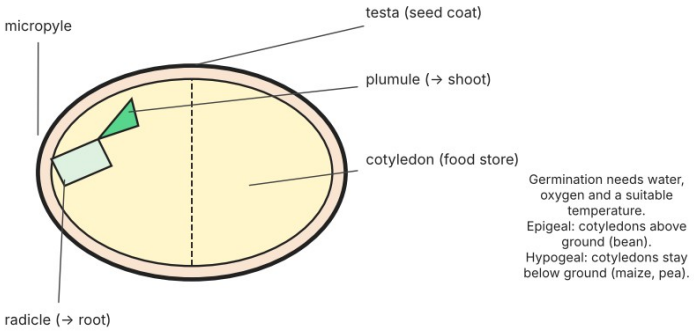
LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ describe seed and fruit formation
- ✔ relate fruit structure to the method of dispersal

Lesson notes

After fertilisation: the zygote → embryo (plumule, radicle, cotyledons); endosperm nucleus → endosperm; integuments → testa (seed coat); ovule → seed; ovary wall → fruit wall (pericarp); ovary → fruit. Petals, stamens and style usually wither.



Method of dispersal	Features of fruit/seed	Examples
Wind	light; wings or hairs (parachutes)	Terminalia, cotton, Tridax, Bombax (kapok)
Animals (eaten)	brightly coloured, juicy, sweet fruits; hard seeds pass through the gut	mango, guava, pawpaw, tomato
Animals (attached)	hooks or sticky surfaces	Bidens (black jack), Desmodium
Water	light, fibrous, waterproof fruits that float	coconut, water lily
Self (explosive)	dry pods split suddenly, throwing seeds	beans, Crotalaria, cowpea, castor oil

Importance of dispersal: reduces competition with the parent and between seedlings; colonises new areas; survival of the species.

Practice Questions: Lesson 5

1. State the method of dispersal of (a) coconut (b) Bidens (black-jack) (c) mango.

Answers: *1. (a) water (b) animals (hooks) (c) animals (eaten)*

Lesson 6: Germination

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ state the conditions for germination
- ✔ distinguish between epigeal and hypogeal germination

Lesson notes

Germination is the development of the embryo into a seedling.

- **Conditions:** water (activates enzymes, softens the testa, swells the seed), oxygen (aerobic respiration), suitable temperature (enzyme activity); some seeds need light or a period of dormancy.
- **Events:** the seed absorbs water (imbibition) → gibberellin is released by the embryo → it stimulates the aleurone layer (in cereals) to make **amylase** and proteases → stored starch → maltose → glucose; proteins → amino acids; fats → fatty acids and glycerol → transported to the growing embryo → respiration increases; the radicle emerges first, then the plumule.
- **Epigeal germination:** cotyledons are carried above the ground (the hypocotyl elongates), e.g. beans, groundnut. **Hypogeal germination:** cotyledons stay below the ground (the epicotyl elongates), e.g. maize, pea.
- During early germination the dry mass decreases (food respired) until the leaves begin photosynthesis.

Practice Questions: Lesson 6

1. State three conditions necessary for germination.
2. Describe the role of gibberellin in the germination of a cereal grain.

Answers: *1. water, oxygen, suitable temperature 2. it stimulates the aleurone layer to make amylase, which breaks down starch in the endosperm to maltose for the embryo*

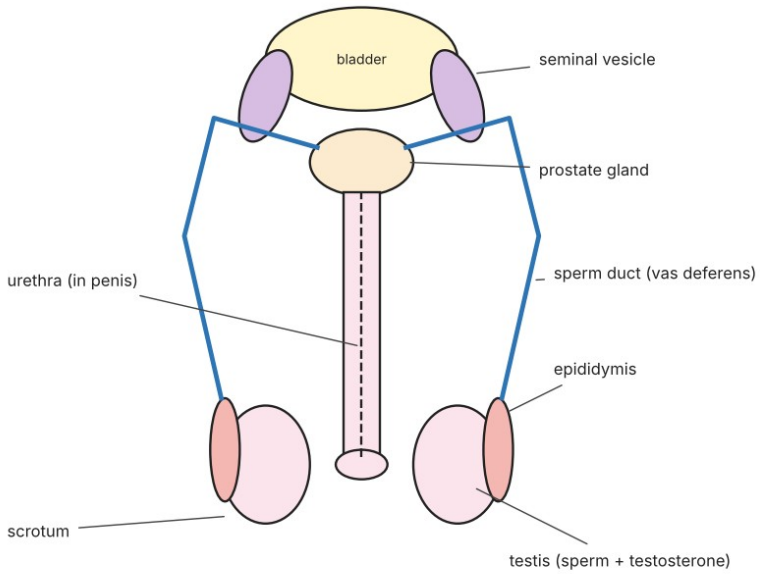
Lesson 7: The male reproductive system

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ describe the male reproductive system and the structure of a sperm

Lesson notes



Part	Function
Testes (in the scrotum)	produce sperm (in seminiferous tubules) and testosterone (by interstitial/Leydig cells); kept about 2–3 °C below body temperature
Epididymis	stores sperm while they mature
Sperm duct (vas deferens)	carries sperm to the urethra
Seminal vesicles	secrete a fluid rich in fructose (energy for sperm)
Prostate gland	secretes an alkaline fluid that neutralises acidity in the urethra and vagina
Cowper's glands	secrete mucus (lubrication)
Urethra	carries semen (and urine, at different times)
Penis	erectile tissue; places semen in the vagina

Practice Questions: Lesson 7

1. State the function of (a) the seminal vesicles (b) the epididymis.

Answers: *1. (a) secrete fluid with fructose for sperm energy (b) stores sperm while they mature*

Lesson 8: Hormonal control of the testes and spermatogenesis

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

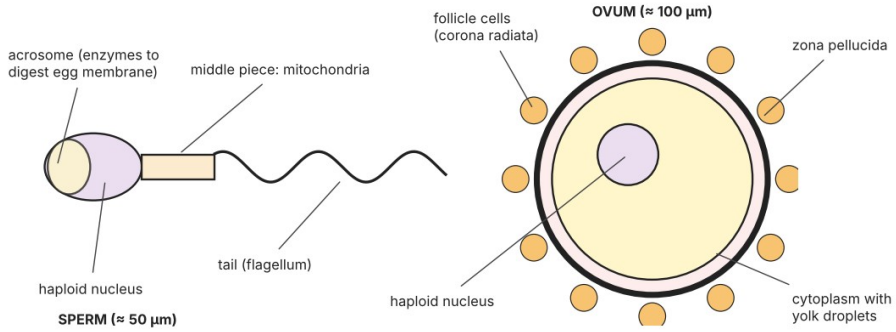
- ✔ describe the hormonal control of the testes
- ✔ describe spermatogenesis

Lesson notes

Hormonal control: the hypothalamus releases GnRH → the anterior pituitary releases **FSH** (stimulates spermatogenesis, via Sertoli cells) and **LH** (stimulates interstitial cells to make **testosterone**). Testosterone promotes sperm production and male secondary sexual characteristics (deep voice, facial and body hair, muscle growth) and inhibits LH release (negative feedback). Inhibin from Sertoli cells inhibits FSH.

Spermatogenesis (in the seminiferous tubules):

- Germinal epithelium cells (2n) divide by mitosis → **spermatogonia** (2n).
- Spermatogonia grow → **primary spermatocytes** (2n).
- Meiosis I → **secondary spermatocytes** (n); meiosis II → **spermatids** (n).
- Spermatids mature into **spermatozoa**, nourished by Sertoli cells.



Practice Questions: Lesson 8

1. Which hormone stimulates the interstitial cells to make testosterone?
2. How many sperm are formed from one primary spermatocyte?

Answers: *1. LH (ICSH) 2. four*

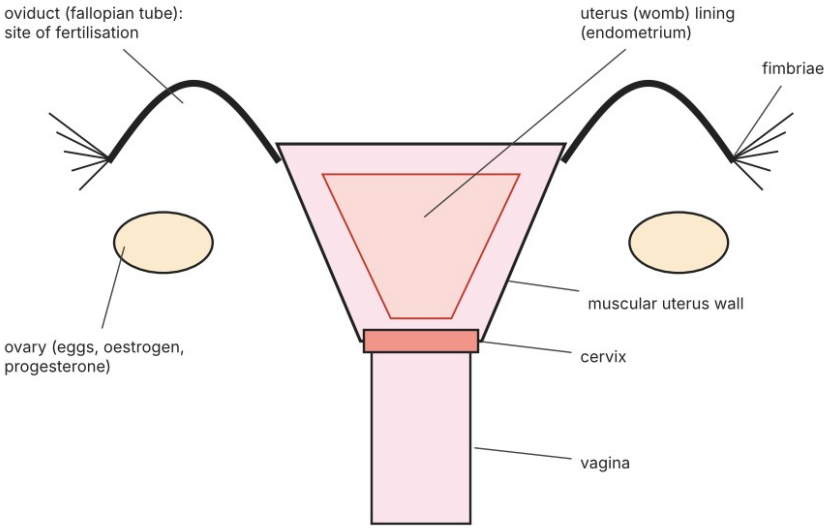
Lesson 9: The female reproductive system

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ describe the female reproductive system

Lesson notes



Part	Function
Ovaries	produce ova (eggs) and the hormones oestrogen and progesterone
Oviducts (fallopian tubes)	carry the ovum to the uterus by cilia and peristalsis; site of fertilisation
Uterus	muscular organ in which the embryo implants and develops; lining = endometrium
Cervix	ring of muscle at the base of the uterus; dilates during birth
Vagina	receives the penis and sperm; birth canal

Practice Questions: Lesson 9

1. Where does fertilisation normally take place?

Answers: *1. in the oviduct (fallopian tube)*

Lesson 10: Hormonal control of the ovaries

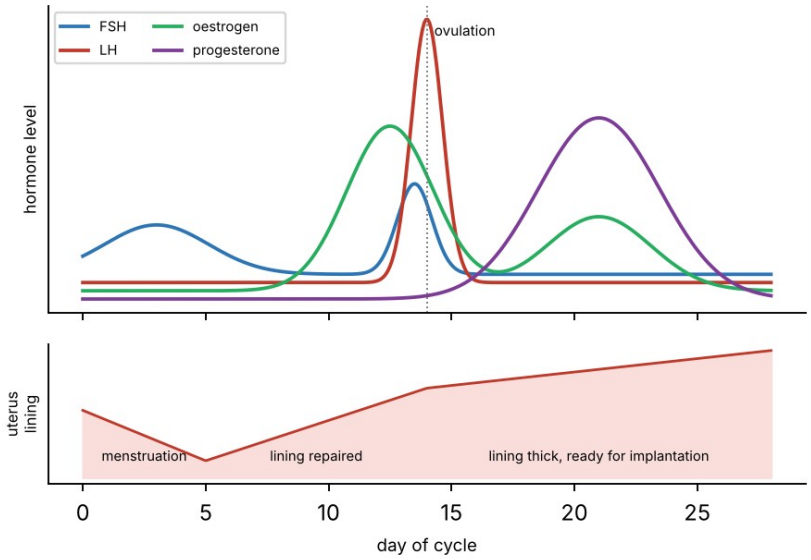
LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ describe the roles of the hormones that control the ovarian cycle

Lesson notes

- The hypothalamus releases GnRH → anterior pituitary releases **FSH** and **LH**.
- **FSH** stimulates a follicle to develop in the ovary; the follicle secretes **oestrogen**.
- **Oestrogen** repairs and thickens the endometrium; at low levels it inhibits FSH (negative feedback), but a high level just before mid-cycle causes a surge of LH (positive feedback).
- The **LH surge** causes **ovulation** (around day 14) and turns the empty follicle into the **corpus luteum**.
- The **corpus luteum** secretes **progesterone** (and some oestrogen), which maintains the thick endometrium and inhibits FSH and LH.
- If there is no pregnancy, the corpus luteum degenerates, progesterone falls, the lining breaks down (menstruation), and FSH rises again: a new cycle starts.



Practice Questions: Lesson 10

1. Which hormone surge causes ovulation?
2. State two effects of progesterone.

Answers: *1. LH 2. maintains the uterine lining; inhibits FSH and LH (stops new follicles developing)*

Lesson 11: Menstruation and oogenesis

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ describe the menstrual cycle and oogenesis

Lesson notes

Menstrual cycle (about 28 days): days 1–5 menstruation (loss of the endometrium and blood); days 6–13 follicular phase (lining repaired, follicle grows); day 14 ovulation; days 15–28 luteal phase (corpus luteum, thick lining).

Oogenesis:

- Before birth, oogonia (2n) divide by mitosis and grow into **primary oocytes** (2n), which start meiosis I and stop in prophase I.
- From puberty, each month one primary oocyte completes meiosis I → a large **secondary oocyte** (n) and a small **first polar body**.
- The secondary oocyte is released at ovulation; meiosis II is completed only if a sperm enters → ovum + second polar body.

Feature	Spermatogenesis	Oogenesis
Gametes per meiosis	4 sperm	1 ovum (+ polar bodies)
Timing	continuous from puberty	starts before birth; one per month from puberty to menopause
Size of gamete	very small, motile	large, non-motile, food store
Meiosis	completed in the testis	meiosis II completed only after fertilisation

Practice Questions: Lesson 11

1. In a 28-day cycle, on about which day does ovulation occur?
2. Give two differences between spermatogenesis and oogenesis.

Answers: *1. day 14 2. oogenesis produces one ovum and polar bodies per meiosis, starts before birth and stops at menopause; spermatogenesis produces four sperm and continues through life*

Lesson 12: Fertilisation and development of the embryo

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ describe fertilisation and early development
- ✔ describe the structure and functions of the placenta

Lesson notes

- Sperm swim through the cervix and uterus to the oviduct; **capacitation** makes them able to fertilise.
- **Acrosome reaction:** the acrosome releases enzymes that digest the zona pellucida; one sperm's membrane fuses with the oocyte's; the **cortical reaction** hardens the zona pellucida so no other sperm enter.
- Meiosis II is completed; the male and female nuclei fuse → diploid **zygote**.
- The zygote divides by mitosis (cleavage) as it moves down the oviduct, forming a ball of cells (**morula**) then a hollow **blastocyst**, which **implants** in the endometrium about 7 days after fertilisation.
- The outer cells form the **placenta** with the mother's tissues; the embryo is surrounded by the **amnion** (amniotic fluid protects against shocks) and joined to the placenta by the **umbilical cord**.
- **hCG**, made by the embryo, keeps the corpus luteum active (detected by pregnancy tests). Later the placenta makes progesterone and oestrogen.

Functions of the placenta: exchange of O_2, CO_2, nutrients and wastes (large surface area of villi, thin barrier, counter-current flow; maternal and fetal blood do not mix); barrier to many pathogens and toxins (but not all: HIV, rubella, alcohol, nicotine can cross); passes antibodies to the fetus (passive immunity); secretes hormones.

Practice Questions: Lesson 12

1. State three functions of the placenta.

Answers: *1. exchange of oxygen, CO_2, nutrients and wastes; production of hormones (hCG, progesterone, oestrogen); barrier to some pathogens; passes antibodies*

Lesson 13: The birth process (parturition)

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ describe the process of birth and its hormonal control

Lesson notes

- At about 40 weeks, progesterone falls and oestrogen rises, making the uterus more sensitive.
- The pituitary releases **oxytocin**, causing contractions of the uterus muscle; contractions push the baby's head against the cervix, stimulating more oxytocin (**positive feedback**), so contractions become stronger and more frequent.
- **Stages of labour:** (1) dilation of the cervix; the amnion bursts ("waters break"); (2) delivery of the baby, head first; (3) delivery of the placenta (afterbirth).
- **Lactation:** prolactin stimulates milk production; oxytocin causes milk ejection when the baby suckles. The first milk (colostrum) is rich in antibodies.

Practice Questions: Lesson 13

1. Explain why the control of birth by oxytocin is an example of positive feedback.

Answers: *1. contractions stimulate more oxytocin release, which causes stronger contractions until the baby is born*

Lesson 14: Reproduction (integration activity)

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ apply knowledge of the menstrual cycle and reproduction to real situations

Worked examples

Example: Situation: A 25-year-old woman has a regular 28-day cycle. Her period started on 1 March. (a) On which day is she most likely to ovulate? (b) Which hormone causes ovulation and which gland secretes it? (c) If she becomes pregnant, why does menstruation stop?

(a) About day 14: 14 March.

(b) LH, from the anterior pituitary gland.

(c) The embryo secretes hCG, which keeps the corpus luteum alive; it (and later the placenta) secretes progesterone, which maintains the endometrium and inhibits FSH and LH.

Practice Questions: Lesson 14

1. A woman's cycle is 30 days long. On which day is she most likely to ovulate?

Answers: *1. about day 16 (14 days before the next period)*

END OF FREE SAMPLE

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